

SYSTEM FOUNDRY - BUSINESS MVP

MakeupLedger

Keep every replacement-session promise visible.

Project	2026-08-15-1604-makeupledger
Customer	Small music, art and language academies
Price	KRW 21,000 per studio per month
Status	PASS - automated tests 10/10

1. Project overview

MakeupLedger records replacement-session credits created by absences and cancellations, then prioritizes commitments by expiry date and outstanding value.

2. Target customer

Small music, art and language academies that run recurring one-to-one or small-group lessons.

3. Problem

Make-up promises are often scattered across chat messages and notebooks. Missed expiries cause customer disputes, while too many unbooked sessions create an invisible delivery obligation.

4. Core features

1) Validated credit issue with sessions, expiry and value. 2) Expiry-risk and liability queue. 3) Session redemption and remaining-balance tracking.

5. Usage

Start the static server, issue a credit whenever a class requires replacement, then use the queue to book urgent learners and redeem one session after delivery.

6. System structure

Browser UI -> validation and expiry/liability engine -> localStorage. No external database, account or paid API is required.

7. Technology

HTML5, CSS, modern JavaScript modules, Node.js built-in test runner, Python static server and Docker.

8. Running screen

EVERY MISSED CLASS HAS A PROMISE.

LOCAL-FIRST · SINGLE STUDIO

MakeupLedger

Keep replacement-session promises visible before they expire.

Open credits
3

Sessions owed
5

Outstanding value
KRW 198,000

Need attention
2

Issue make-up credit

Reference

Learner / course

Issue / expiry

Sessions / value

SAVE CREDIT

Use a non-sensitive learner label.

FULFILMENT QUEUE

Promises that need a booking

Expired and near-expiry first

URGENT	Minseo Kim · MK-1042 Piano intermediate · expires 2026-08-18	2 sessions KRW 90,000	3d left
WATCH	Jiwon Park · MK-1047 Watercolor basics · expires 2026-08-29	2 sessions KRW 76,000	14d left
AVAILABLE	Hyeon Lee · MK-1051 English conversation · expires 2026-09-20	1 session KRW 32,000	36d left
CLOSED	Sora Choi · MK-1033 Violin beginner · expires 2026-08-10	0 sessions Settled	Settled

Rendered from the same calculation engine verified by automated tests.

Application state rendered from the verified production calculation engine.

9. Test result

Automated tests 10/10 passed. Coverage includes normalization, invalid dates, remaining sessions, liability, day counts, urgent and overdue classification, totals, duplicate references and final redemption. HTTP start, README commands and the calculation workflow were also verified.

10. Security and limitations

Inputs are length-, range-, date- and status-validated; UI output is escaped. Operators should use non-sensitive learner labels and never enter contact, payment, medical or authentication data. The MVP does not send reminders, book calendars or interpret refund law.

11. Price and revenue model

KRW 21,000 monthly per studio. The value proposition is fewer disputes, fewer forgotten promises and a clear view of outstanding teaching obligations.

12. First-customer plan

Offer ten local lesson studios a free one-week migration of their current make-up list. Measure expired credits recovered and admin time saved, then convert useful pilots to a 14-day paid trial.

13. Improvement plan

1) Calendar booking. 2) Customer reminder links. 3) Policy templates. 4) Teacher assignment. 5) Audit trail and CSV export. 6) Multi-user login.

14. Overall evaluation

Evaluation	Score	Reason
Problem intensity	9/10	Missed commitments can directly cause refunds and disputes.
One-day completeness	9/10	Issue, queue, totals and redemption are complete.
Sales potential	8/10	Studios can compare the fee with prevented refunds.
Maintenance ease	9/10	Static app with no external service.
Differentiation	8/10	Expiry risk and obligation value share one queue.
Expansion potential	8/10	Calendar, reminders and team roles fit naturally.

Overall score 8.5/10

A recurring and financially visible operational problem is reduced to a narrow queue. Before paid release, validate reminder and calendar priorities with real studio owners.

Appendix - verification checklist

PASS: zero external dependencies; program start; core workflow; 10 automated tests; README commands; credential scan; PDF parsing and all-page rendering.

Release decision

PASS - suitable for local operations and customer interviews. Studio policy and final lesson scheduling remain the operator's responsibility.